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# An economic model for the content market on algorithmic social networks

A theory of timing in the dispute for attention: latent demand, active supply and the optimal publication window.

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WRITTEN BY

Francisco Figueiredo · Lauanny Brandão · Pedro Bordinhão

RIO DE JANEIRO · 2026



# An INDICA publication.

Founded in 2024, INDICA is a communications and influencer-marketing agency.

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|---------|-------------------------------------------------------|
| CONTACT | indica.osc@gmail.com · @indica.cc · agenciaindica.com |
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## HOW TO CITE

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# What this paper proposes

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This paper proposes an economic model for interpreting content performance on algorithmic social networks. Treating attention as a scarce resource and content units as supply items competing for limited shares of it, the model articulates topic, format, and source as axes of demand, with weights that vary along the curve. It distinguishes latent from realized demand and published from active supply, showing that saturation results from a cumulative stock of content in algorithmic circulation, not merely from the flow of new publications. It introduces the first and second derivatives of the demand curve as tools for identifying optimal publication timing and derives the closure condition of the optimal window — the equality between the proportional growth rates of latent demand and active supply. It decomposes reaction time into decision time and production time, showing that the strategic reduction of lag comes from anticipating decision, not from anticipating production. The paper discusses macro and micro saturation, the source's base attention, the formation of hybrid demand curves, and strategic implications for the intertemporal allocation of resources. It argues that campaigns paid per contracted piece should maximize average reach per piece, not aggregate topic reach, which shifts the optimal timing to phases preceding peak public visibility. The model offers a unified framework for explaining why similar content produces markedly different results when entering distinct conjunctures of public interest and competition.

## KEYWORDS

attention economy

algorithmic social networks

timing

cumulative saturation

intertemporal allocation

reaction time

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## SECTION 1

# Introduction



Digital communication has become one of the main arenas of political, institutional, economic, and cultural dispute of the present. The accelerated digitalization of society has caused communicational logic to overflow the strict field of media and begin to reorganize politics, the economy, and culture, reconfiguring relations of power and of market in the contemporary environment (Hjarvard, 2013; Piaia et al., 2023). Companies, governments, civil society organizations, multilateral bodies, political movements, press outlets, and independent creators now operate in an environment in which producing content is no longer enough.

The central problem has become the conversion of messages into attention within an ecosystem marked by excess supply, permanent competition, and rapid oscillations of public interest. In information-rich environments, the abundance of messages produces a scarcity of attention capacity, and not the other way around (Davenport & Beck, 2001; Goldhaber, 1997; Simon, 1971; Wu, 2016) – and longitudinal evidence indicates that collective attention cycles have been shortening as the volume of content production and consumption grows (Lorenz-Spreen et al., 2019). The formal economics of this dispute has already been modeled in terms of competition among media for the consumer's limited attention, in which the entry of new sources endogenously reorganizes the circulation conditions of the others (Anderson & de Palma, 2012). Communication in digital environments has thus come to approximate what may be called an attention market. In it, content items function as symbolic goods that compete with one another, audiences represent demand, and the platforms' algorithmic systems operate as mechanisms that distribute visibility and modulate, opaquely, the potential reach of content (Bucher, 2018).

This transformation has altered the conditions under which communication is planned, executed, and evaluated. Users' available time is limited, their cognitive disposition is finite, and the volume of stimuli in circulation far exceeds their capacity for absorption. Content items compete for restricted shares of interest, visibility, engagement, and permanence. Digital communication thereby takes on characteristics proper to a market in which different agents compete for the allocation of a scarce resource. The analogy with traditional markets offers an important starting point, because it makes it possible to understand that there is a relation between the quantity of content offered on a topic and the interest the public manifests in that topic in a given period. From this relation derive variations in the visibility each content item obtains, which warrants treating organic reach as the main empirical expression of the result that distinct content items manage to obtain in specific conjunctures of the dispute for attention.

This framing gains particular relevance in view of the fact that a large share of decisions in digital communication is still made without an analytical model capable of organizing, in an integrated manner, public interest, content production, timing, saturation, and marginal return. There is an abundance of metrics, reports, and performance indicators, but this does not by itself produce a consistent understanding of the economic structure that conditions those results. Frequently, content performance continues to be explained in fragmented fashion, now by its intrinsic quality, now by the strength of the source, now by algorithmic changes, now by conjunctural factors treated as isolated episodes. The result is an environment in which campaigns, institutions, and strategies operate with a great volume of data and a low capacity to interpret the ecosystem in which they compete for attention.

It was in this context that social organizations, movements, and collectives working on socio-environmental and political causes began to incorporate digital influencers into their strategies, seeking to harness these creators' symbolic capital, credibility, and reach to amplify the visibility of mobilization, advocacy, and public-engagement campaigns (Goodwin et al., 2023). Even with the growing diffusion of these strategies, results frequently fall short of their potential. The problem is broader than the specific case of influencer campaigns: it points to the absence of a formulation capable of explaining why content and campaigns with similar resources, relevant topics, and technically consistent execution exhibit such distinct performances.

The formulation developed here starts from the idea that these performance differences depend on four articulated variables. First, the relative position of content items on the demand curve for the topic they address. Second, the speed with which that demand shifts. Third, the simultaneous expansion of competing supply — including the cumulative stock of previously published content that remains in algorithmic circulation. Fourth, the capacity different agents have to react to these changes. The decisive point lies less in the isolated attributes of each piece and more in its temporal and relational insertion into a competitive ecosystem of attention (Citton, 2017). The performance of a content item thus comes to depend on its position within a specific conjuncture, in which demand for a topic may be accelerating, stabilizing, or declining, while supply responds to these movements with greater or lesser speed.

This shift requires a more precise treatment of the elements that compose the model. Digital content items can be understood as supply units whose valuation is determined by a combination of fundamental components. Among them, the topic addressed, the format used, and the person or instance that communicates stand out. The topic occupies a central place in this composition because it concentrates the most abrupt movements of expansion and retraction of public interest, becoming the main strategic axis for understanding performance oscillations in the digital content market. The format affects the capture of attention and the adaptation to the platform's language. The source weighs on credibility, on the capacity to activate niches, and on the conversion of diffuse interest into effective engagement. What is proposed here is to bring these dimensions together in a single model, preserving their articulation and the unequal weight they exert on demand.

The notion of the derivative of the demand curve occupies, in this context, a central role. The moment of greatest communicational efficiency does not necessarily coincide with the absolute peak of interest in a topic. In many cases, it lies in the period in which demand accelerates most intensely. The model therefore draws attention to the rate of variation of interest, and not only to its already consolidated volume. This shifts the analysis to the terrain of timing and makes it possible to understand why content published on the same topic, with similar resources and comparable quality, can achieve radically distinct results depending on the stage of the curve at which it enters circulation.

It is worth demarcating, from the outset, the scope of this proposal. To sustain the centrality of timing is not to claim that the moment of publication is the most important determinant of a content item's performance. The essence of the piece, its quality, its execution, and its adherence to the audience probably weigh more than the instant at which it enters circulation. The model concentrates on timing for three reasons of distinct order. First, when one observes a set of content items, and not an isolated piece, timing is the only variable that is effectively common and isolable across them, whereas quality, format, and source vary idiosyncratically and in ways that are hard to compare. Second, it is the variable easiest and cheapest to manipulate: the same campaign, with exactly the same content, can obtain a markedly more expressive result merely through the choice of the moment of publication. Third, in contracted campaigns, the organization already optimizes the other variables when selecting sources that combine base attention, an audience adherent to the topic, wide reach, and good production quality; once these dimensions are maximized in the choice of

sources, timing remains the decisive lever still under its direct control, and there would be no reason to leave it unexplored. The intention, therefore, is not to model a content item's reach from its form, an unfeasible task given unobservable and hardly parameterizable variables such as quality; the focus falls on the dimension the agent can, in fact, adjust.

To this dynamic is added the notion of reaction time, understood as the interval between the identification of a communicational opportunity, the decision to act, and the effective publication of the content. In digital markets, this interval acquires great importance because the demand curve for certain topics can change rapidly, so that small differences in time produce expressive impacts on the reach obtained. Agents with shorter reaction times tend to position their content in phases of demand acceleration, when the marginal return tends to be higher. Agents with longer reaction times enter more frequently into already congested environments, in which attention is more heavily disputed and the effort required to convert message into reach grows considerably. Time thus becomes part of the very formation of communicational value.

The usefulness of a model such as this one goes beyond the interpretation of performance on social networks. It offers a key for thinking about how different organizations allocate communicational resources in contexts of scarce attention. This matters to governments that need to communicate public policies, to social organizations operating on restricted budgets, to multilateral institutions that depend on the public diffusion of complex agendas, to companies seeking to insert topics or products into rising conversations, and to researchers interested in understanding the political economy of content circulation. In all these cases, the underlying question is similar: how to make better decisions in an environment in which attention, visibility, and timing have become strategic assets.

The central contribution of this paper lies in articulating, within a single framework, seven elements: (i) the distinction between latent demand and realized demand, which confronts the methodological circularity of measuring interest in a system mediated by algorithms; (ii) the distinction between published supply and active supply, which recognizes the cumulative character of saturation; (iii) the objective function of campaigns contracted on a per-piece basis, which maximizes average reach per piece and not the aggregate reach of the topic; (iv) the use of the first and second derivatives of the demand curve as timing signals, culminating in the formal closure condition of the optimal window; (v) reaction time decomposed into decision time and production time, showing that the strategic reduction of lag comes from anticipating the decision; (vi) the source's internal exhaustion curve, which explains why the source's strength does not by itself sustain performance in late phases; and (vii) hybrid curves as a strategy of thematic recombination. It is argued, in sum, that it is possible to develop a more rigorous language for interpreting the dynamics of digital communication and for guiding, on more consistent grounds, strategic decisions of production, publication, and resource allocation.

## SECTION 2

# Supply, demand, and reach in the digital content market

## 2.1 The market analogy

In analogy with traditional markets for goods and services, there is a relation between the quantity of content offered on a given topic and the interest the public manifests in that topic in a given period. From this relation derive variations in the visibility each content item obtains, which makes it possible to treat organic reach — the volume of views and circulation obtained without paid promotion — as the main observable variable of the outcome of the dispute for attention. The usefulness of this formulation lies in allowing content circulation to be observed as a competitive process, in which different agents dispute a scarce resource and obtain distinct returns according to the relation among social interest, the volume of competing publications, and the moment of entry into the communicational flow.

The platforms that organize this circulation can be understood as infrastructures that structure interactions among different groups of agents, articulating economic, social, and algorithmic logics within a single architecture of visibility (van Dijck et al., 2018). In this attention market, reach should not be understood as analogous to price — which, in traditional markets, operates as the adjustment variable that balances supply and demand — but as the quantity transacted, that is, the volume of attention effectively converted into circulation. The implicit “price” corresponds to the opportunity cost (productive effort, time, symbolic capital, entry window) that each agent pays to dispute a place in algorithmic distribution.

Attention markets have a long historical genealogy. Wu (2016) describes how, over the course of the twentieth century, sensationalist newspapers, commercial radio, broadcast television, search engines, and social networks successively consolidated themselves as economic infrastructures whose central function is to capture public attention and resell it to advertisers or contractors. The current digital ecosystem inherits this logic and recomposes it under a new architecture: the capture of attention is decentralized across millions of creators and profiles, intermediation is mediated by opaque algorithms, and the sale of that attention, formerly carried out by single structures (a TV network, a newspaper), comes to operate in overlapping layers — platforms, agencies, creators, contractors. What changes is not the nature of the market, but its granularity and its mechanisms of mediation. In this new arrangement, digital content can be characterized as a contingent commodity (Nieborg & Poell, 2018): its format, reach, and durability are not fixed properties, but results continuously produced by the platform’s governance rules, which can change unilaterally and with little prior notice.

The formal reference closest to what is proposed here is the model of competition for attention of Anderson and de Palma (2012), in which sources dispute consumers’ limited attention and the entry of new competitors endogenously reorganizes the circulation conditions of the others. The present model starts from that result, but shifts the analytical axis in three directions: it treats the dispute as a dynamic trajectory, and not as a static equilibrium; it models competition as a cumulative stock with decay, and not as a set of simultaneous sources; and it adopts as its objective function the average reach per contracted piece, relevant for campaigns paid on a per-piece basis (section 2.5). By dispensing with the search for a static equilibrium point, typical of simpler Walrasian formulations<sup>1</sup>, the model examines supply and demand as continuous temporal trajectories

without direct comparability of measures. The analytical axis does not fall on the static crossing of these curves, but on the speed and the relative distance between them.

## 2.2 Latent demand and realized demand

Treating demand as a directly observable variable would be an analytical error in digital environments. The platforms' recommendation mechanisms operate on the basis of attention signals — viewing time, retention, engagement — and the best quantitative clue available about interest in a topic is, strictly speaking, the performance of content on that topic within the system itself. It so happens that this clue is simultaneously the dependent variable of the analysis: performance is what one wants to explain and maximize. There is, therefore, a methodological circularity that needs to be confronted explicitly.

To this end, a distinction is drawn between latent demand and realized demand. Latent demand, denoted  $D_L(t)$ , corresponds to the potential social interest in the topic at a given instant: there exists an aggregate curve of public relevance that could manifest itself in observable behavior, given adequate conditions of supply, format, source, and algorithmic mediation. Realized demand, denoted  $D_R(t)$ , corresponds to the share of that latent demand that effectively converts into observable behavior: viewing, retention, sharing, commenting, searching, clicking, engagement. By construction,  $D_R(t) \leq D_L(t)$ , and the difference between the two is a function of the conditions of competing supply, of the quality of algorithmic mediation, and of the capacity of the sources present to activate the existing demand.

This distinction is central because it prevents a naive reading of circulation metrics. A content item may perform poorly not for lack of latent demand, but because it entered late, because the source had no adherence to the topic, because the format did not capture attention, or because competing supply had already occupied the available window. At the same time, when multiple different content items, from varied sources, perform above average on the same topic within a short interval of time, the signal points with reasonable confidence to high realized demand — although it does not permit directly inferring the size of the underlying latent demand.

From an empirical standpoint, the estimation of  $D_L$  requires triangulation. Indicators that approximate latent demand, because they are less sensitive to the competitive structure internal to the platforms, include search volume in search engines, the volume of mentions in open environments, the intensity of journalistic coverage, and qualitative conversational signals. Indicators that approximate realized demand include the aggregate reach and engagement of content on the topic. Each set has distinct structural biases: the former underestimates attention mobilized exclusively inside closed platforms; the latter are contaminated by the very supply whose relation to demand one wants to understand. Good practice is to combine sources and treat the demand curve as an operational approximation, not as an exact measure.

## 2.3 Digital content as an information good

From an economic standpoint, digital content is an information good of non-rival nature and near-zero marginal cost of reproduction (Shapiro & Varian, 1999). The same content item can be consumed simultaneously by an indefinite number of users without one individual's consumption reducing availability for others, distinguishing it from the rival goods typical of traditional markets. This non-rivalry does not eliminate scarcity: scarcity shifts from the good itself to the attention required for its enjoyment. In digital environments, content does not compete for material access, but for time, cognitive focus, and algorithmic visibility.

The abundance of information goods does not dissolve competition. It merely reorganizes that competition around the capacity to transform available content into effective attention. This conversion is mediated by recommendation systems that hierarchize circulation, amplifying some signals, compressing others, and redistributing visibility in opaque ways (Bucher, 2018; Gillespie, 2014). Algorithmic mediation introduces strong nonlinearity: small initial differences in retention, interaction, or speed of response can generate much larger differences in distribution at later stages. Experimental evidence from cultural markets shows that, under social influence, small initial differences generate success trajectories that are highly distinct and partially independent of the intrinsic quality of the content (Salganik et al., 2006).

## 2.4 Published supply and active supply: the cumulative character of saturation

In traditional markets, supply can be treated as a flow of goods available in a given period. On algorithmic social networks, this formulation is insufficient. A content item published on a given day does not automatically exit the market the following day: it continues to be recommended, may gain second waves of circulation, may appear to new audiences days later, and goes on competing with new publications for the same available attention.

A distinction is therefore drawn between published supply and active supply. Published supply,  $S_p(t)$ , corresponds to the volume of new content on the topic entering the system at instant  $t$  (a flow variable). Active supply,  $S_A(t)$ , corresponds to the set of content items that continue to receive algorithmic distribution at  $t$ , regardless of the instant at which they were published (a stock variable, with decay). In formal terms,  $S_A$  can be represented as the accumulated integral of  $S_p$  over a window of decreasing algorithmic relevance.

Explicitly, letting  $w(a)$  be the decay kernel — the fraction of algorithmic relevance a piece retains after time  $a$  has elapsed, with  $w(0) = 1$  and  $w$  decreasing —, we have

$$S_A(t) = \int_{-\infty}^t S_p(u) w(t - u) du,$$

where:

$S_A(t)$  — active supply at instant  $t$ : the stock of content on the topic that still circulates and competes for attention at that moment;

$S_p(u)$  — published supply at instant  $u$ : the flow of new content that entered the system at that moment in the past;

$w(t - u)$  — decay kernel applied to the piece's age: the fraction of algorithmic relevance that a content item published at  $u$  still retains when instant  $t$  arrives; it equals 1 at the moment of publication and diminishes with age;

$\int_{-\infty}^t \dots du$  — continuous sum over all past instants  $u$  up to the present  $t$ : each moment of the past contributes what it published, discounted by decay.

In words: today's active supply is the sum of everything already published on the topic, with each piece counting in proportion to how much it still circulates. Each content item has a half-life  $t_{1/2}$  — the age at which  $w(t_{1/2}) = 1/2$  —, which measures how quickly it ceases to circulate; its value depends on the platform and the format and must be estimated empirically in each case. Without decay ( $w \equiv 1$ ), the integral recovers the total already published, a stock that only grows; with decay,  $S_A$  is a moving stock, which advances while there is

new publication and recedes when the flow diminishes. It is this form — an integral, and not an instantaneous count — that grounds the cumulative character discussed below. The literature on the dynamics of collective attention gives empirical support to this representation: the decay of attention to individual items follows regular and measurable patterns (Wu & Huberman, 2007), well described by composite decays with a phase of active diffusion and another of residual memory (Candia et al., 2019), and items of social circulation exhibit slow relaxation and second waves after exposure peaks (Crane & Sornette, 2008) — precisely the stock-like behavior that the notion of active supply is meant to capture.

The distinction matters because saturation is a function of active supply, not of published supply. At the peak of the demand curve, competing supply is not merely elevated in flow, because more agents are publishing simultaneously — it is also the largest in stock, because everything published in the preceding days is still circulating. This cumulativeness significantly amplifies the saturation effect at the peak and reinforces the argument, developed in section 4, that the peak of public visibility is not the point of maximum communicational efficiency.

It should also be noted that distinct content items have very unequal capacities to capture attention. A publication by a highly legitimized source, in a widely replicable format, with strong adherence to the platform's language, does not weigh the same as a peripheral post. Active supply must therefore be understood as a heterogeneous set, weighted by the competitive weight of each piece that composes it. Formally, it suffices to read  $S_p(u)$  not as a count, but as the sum of the competitive weights  $\kappa_j$  of the pieces published at  $u$  — the integral then accumulates effective supply, and not gross volume. The operational definition of these weights (by source, format, and adherence) is an empirical task prior to any test of the model, taken up again in section 8; the heterogeneity they capture is discussed in section 3.

## 2.5 Objective function: what is being maximized

The formulation of the model requires making explicit which variable is being maximized. This decision has important practical consequences and is not always made explicit in analyses of digital communication.

For society or the public debate in a broad sense, a topic's peak is the moment when there is the greatest aggregate public conversation about it. For a campaign that pays a unit price per contracted content item, the problem is distinct: the objective is to maximize the average reach of the financed pieces, not the summed reach of all circulation of the topic in the ecosystem. When a campaign contracts a fixed set of content items under a given budget and pays a unit price per piece, the number of pieces is defined a priori. The remaining decision variables — when to publish, with which source, in which format, under which framing — must be optimized with reference to the expected average reach per piece, and not to the sum of the topic's circulation in the ecosystem, which also includes content not contracted by the campaign. It is additionally assumed that the reaches of the contracted pieces are approximately independent of one another; when there are relevant synergies among pieces of the same campaign — cross-amplification, narrative sequencing —, the timing calculation must incorporate them. The consequence of this objective function for the optimal moment of publication is developed in section 4: the best moment for the campaign does not coincide, in general, with the moment of greatest aggregate visibility of the topic.

It must be acknowledged, however, that not all communication has the maximization of average reach as its sole objective. There are symbolic, political, and institutional reasons for organizations to mark public presence at moments of high visibility — dialoguing with institutional actors, rendering accounts, demonstrating a position, fulfilling a specific political function. The model does not claim that publishing at the peak is always wrong. It states something more precise: if the main objective is to maximize average reach per contracted

piece, the peak tends to be suboptimal. When there are other objectives — legitimacy, political pressure, institutional presence, symbolic visibility —, it may make sense to consciously accept lower reach efficiency in exchange for other gains. The problem arises when this choice is made without clarity, on the assumption that publication at the moment of greatest aggregate visibility is also the moment of greatest communicational efficiency.

## 2.6 The relational rule

The distinctions introduced make it possible to formulate the model's central relational rule. For a campaign contracted on a per-piece basis, the expected reach of each contracted piece, at the instant of its publication, tends to be proportional to the ratio between latent demand and competing active supply at that instant:

$$\text{expected reach per piece} \propto D_L(t) / S_A(t),$$

where:

$D_L(t)$  — latent demand: the public's potential interest in the topic at the instant  $t$  of publication (section 2.2);

$S_A(t)$  — active supply: the stock of competing content still in circulation at that instant, weighted by the competitive weight of each piece (section 2.4);

$\propto$  — proportionality symbol: expected reach rises and falls together with the ratio, without commitment to an exact numerical value.

The intuition of the formula is direct. The numerator represents the attention potentially available for the topic; the denominator represents the set of content items disputing that attention. When latent demand grows faster than active supply, the ratio rises and each new piece finds ample relative space. When active supply grows faster than latent demand, or when latent demand loses momentum while active supply remains elevated, the ratio falls and each new piece disputes an ever smaller fraction of the available attention. It is worth specifying that  $S_A$ , in this ratio, does not designate the mere count of competing content items, but active supply weighted by the competitive weight of each piece, in the sense introduced in 2.4: what presses the ratio downward is effective competing supply — weighted by source, format, and adherence —, not the gross volume of publications. The optimal instant of publication is, in consequence, the one at which this ratio is at its maximum — a point taken up again in section 4, where the derivatives of the demand curve function as anticipatory signals of its opening and its closing.

The form of the rule can be partially microfounded. If the attention effectively realized on the topic is divided among the pieces of the active supply in proportion to their competitive weights, the expected reach of a piece of weight  $\kappa_i$  is of the order of  $D_R(t) \cdot \kappa_i / S_A(t)$ . Substituting  $D_L$  for  $D_R$  in the numerator presupposes that the conversion rate between latent and realized demand is approximately stable within the decision window — a working hypothesis, not a result, since section 2.2 itself recognizes that this conversion depends on supply conditions. Two additional simplifications deserve note. First, simple proportionality implies unit elasticities of reach with respect to numerator and denominator; a generalization of the form  $D_L^\alpha / S_A^\beta$ , with  $\alpha, \beta > 0$ , would preserve the qualitative argument — the existence of an optimal window prior to the peak — altering only its exact position. Second, the published piece is itself stock: strictly speaking, its total accumulated reach

corresponds to the integral of the ratio over its useful life, weighted by its decay; evaluating the ratio at the instant of publication is a first-order approximation, the better the shorter the piece's half-life relative to the scale of variation of the demand curve.

It is worth registering, accordingly, that the formulation operates as an operational approximation, not as an exact measure. Since  $D_L$  is estimated through partial proxies and  $S_A$  is in turn a function of the same dynamics one wants to understand, the rule should be read as a guide for reasoning under structural uncertainty — useful for distinguishing qualitatively distinct phases of the curve and for orienting allocative decisions, even though it does not permit computing with precision the expected reach at each instant. The circularity introduced in section 2.2 does not disappear through the formulation of the rule; it is merely rendered manageable.

The synthesizing formulation can be stated as follows: the demand relevant to a campaign is not merely the aggregate interest in the topic, but the share of that interest not yet captured by competing active supply. What matters is the attention available, not merely the attention in existence.

## 2.7 Supply as an endogenous variable

The supply curve has an endogenous character: it is not a passive stock, but grows as an immediate reaction to the heating of demand. Pioneering publications function as endogenous shocks that organize public interest and expand the topic's own demand. But the high performance of these first content items rapidly attracts new producers. A rapid ecosystemic transition is established: the initial expansion tends to generate competitive congestion that erodes the marginal return and alters the incentives of late publications. This endogenous character is amplified by platformized governance (Nieborg & Poell, 2018): the very rules of algorithmic circulation — which signals the system amplifies, which formats it privileges, how long a content item remains in distribution — are produced and modified by the platforms, which makes supply dynamics sensitive not only to agents' behavior, but to governance decisions beyond their control.

This competitive expansion occurs asymmetrically. Smaller profiles with greater operational autonomy tend to react quickly, while institutional agents face higher temporal costs of coordination. The rapid, decentralized entry of new publications alters the conditions of the ecosystem, intensifying competitive pressure and reducing marginal reach even before slower sources manage to enter the market. When the organic expansion of supply occurs at the exact moment of demand acceleration, it exerts a multiplicative effect on the level of saturation. This is a saturating spillover, in which spontaneous supply negatively affects the marginal performance of content produced by agents operating with longer reaction times. The digital content market is thus traversed by recurrent competitive externalities: one actor's publication can alter the circulation conditions of the others.

There exists an inflection point beyond which the addition of new content ceases to produce demand-expanding effects and begins to operate as a saturating force. This point corresponds to the transition between the phase of accelerated growth and the phase of stabilization of the curve, the moment at which active supply becomes large enough that additional supply begins to compete for attention rather than expand it. From that point on, the market enters another configuration: the public has already been widely exposed to the topic, the differential between content items tends to diminish, and the additional attention each new piece manages to capture starts to fall more sharply.

It is worth making explicit, finally, the equilibrium status of these formulations. The model operates in partial equilibrium: it takes the trajectory of competing supply as given and asks what the best moment of entry is for an individual agent. If all agents simultaneously adopted the anticipation rule that follows from section 4,

active supply itself would shift toward earlier phases of the curve and the window of maximum ratio would migrate — in the limit, the informational advantage would be dissipated by competition, as in any arbitrage. What preserves, in practice, the rent of anticipation is the structural heterogeneity of reaction times postulated in section 5: as long as coordination costs keep a relevant share of agents unable to respond within the window, anticipation remains advantageous for those who can exercise it. From this follows, moreover, an observable implication, registered in the research agenda of section 8: in more professionalized topic markets, the average lead time of publications relative to the peak is expected to increase over time.

## SECTION 3

# Structural components of demand: topic, format, and source



The dynamics of digital content can be decomposed into its internal components. Topic, format, and source participate simultaneously in the structure of supply and in the structure of demand. On the supply side, they define what is put into circulation, by whom, in what form, and around which theme. On the demand side, they condition the public's differential attention to particular topics, formal arrangements, and specific voices. In this model, these components are treated primarily as axes of demand, because it is the dynamics of the latter that occupies the central position in explaining performance oscillations. This analytical choice does not eliminate their role in the constitution of supply, but organizes the model from the side of the relation that best permits understanding the variation of reach.

## 3.1 Topic

The topic is the central theme addressed by the content. Examples include intergovernmental conferences, environmental disasters, elections, specific public policies, denunciations of rights violations, and other subjects that gain public relevance at particular moments. The topic is highly sensitive to historical context, news cycles, crises, and institutional agendas, which makes its demand oscillate with significant frequency — behavior typical of high thematic volatility and exposure to exogenous shocks, long described, at the qualitative level, by the issue-attention cycle (Downs, 1972). In some cases, the oscillation is abrupt and concentrated in short periods. In others, traction extends over a longer time, though at variable rhythms. This centrality stems from the fact that the topic concentrates the most intense movements of expansion and retraction of public interest, functioning as the main vector of displacement of the demand curve.

## 3.2 Framing and format

Between topic and format operates a decisive intermediate dimension: framing. In communicational terms, to frame is to select certain aspects of reality and make them more salient, producing competing versions and interpretive keys of the same topic (Entman, 1993). The format, in turn, transcends the media category (video, text, audio) to define the narrative and aesthetic structuring of the piece. The format is the gear that dialogues directly with algorithmic logics and with the consumption preferences of each platform, determining the initial capture of attention, the intelligibility of the content, and its capacity to propel large-scale imitative dynamics.

In general, format trends vary at a less accelerated rhythm than interest in the topic. This does not mean the format is secondary or stable in a rigid sense. In certain contexts, a specific format can come into evidence very rapidly and produce its own valuation cycle, especially when its replicability is high and its adherence to the platform's language is particularly strong. Even in these cases, the format remains conditioned by a relation with the topic and with the source: not every format in evidence is adequate to any type of topic, especially in the case of content that requires greater contextualization or another regime of attention.

### 3.3 Source and base attention

The source represents the person, institution, profile, or instance that conveys the content, with its set of attributes — reputation, symbolic capital, niche of activity, relationship with the audience, communicative style, and history of positioning. Demand for particular sources can be relatively stable, insofar as their audiences remain interested in following them, regardless of occasional variations in the topics addressed.

This stability configures what is denominated, from here on, the source's base attention: a relatively stable capacity to mobilize attention that does not derive solely from the demand curve of the topic addressed, but from the already consolidated relationship with a specific community of audience. In consequence, content on cold topics can perform well when published by sources with strong adherence to a given niche, whereas content on hot topics can yield less than expected when conveyed by instances whose connection to that interpretive universe is weak or barely credible.

Whenever the role of the source is examined, the niche to which it belongs must be considered simultaneously. In digital environments, profiles do not operate in isolation: they represent micro-spaces of consumption, interpretive micro-communities, and micro-structures of demand. For this reason, the source is, at the same time, a gateway to a specific micro-structure of demand. This observation prepares the discussion, taken up again in section 6, of how this micro-structure has its own dynamics of interest, tolerance of repetition, and thematic exhaustion, and thus introduces an intermediate layer between the general demand for a topic and the concrete performance of a content item.

### 3.4 Variable hierarchy among the components

From this decomposition, it becomes possible to state that, although the three components are relevant, the topic exhibits the greatest degree of volatility and, in consequence, tends to become the most sensitive strategic variable for understanding performance oscillations in the digital content market. In situations in which a given topic gains traction rapidly, content on that topic tends to perform better, provided it is published within the adequate interval, taking advantage of phases of accelerated demand expansion, in which the marginal return increases. This primacy of the topic does not eliminate the importance of format and source. It indicates, rather, that the most abrupt movements of valuation and devaluation of demand tend to concentrate in the thematic trajectory: the topic opens windows, accelerates interest, produces inflections, and reorganizes the competitive environment with greater speed than that observed, in general, in format preferences or in average loyalty to a source. It should be registered that this is a structural postulate of the model — plausible in light of the attention cycle described by Downs (1972), but not yet tested comparatively across the three dimensions; section 8 converts it into an explicit hypothesis.

The format, in turn, plays a decisive role in the way this demand is captured, intensified, or converted into effective circulation. When a specific form of presentation gains collective adherence and comes to be replicated en masse, it can significantly increase a content item's capacity for penetration, including by making more accessible topics that, in other formal configurations, would circulate with more difficulty. There are moments in which the emergence of a replicable format alters the frontier of the possible for a topic, expanding its capillarity and reorganizing the dispute for attention. Even so, the topic preserves analytical centrality, while the format frequently operates as a mediator of the form in which interest materializes into consumption, sharing, and retention.

The source completes this structure by functioning as an anchoring point of trust, familiarity, and entry into specific niches. The strength of the source does not depend only on the size of the audience: it also de-

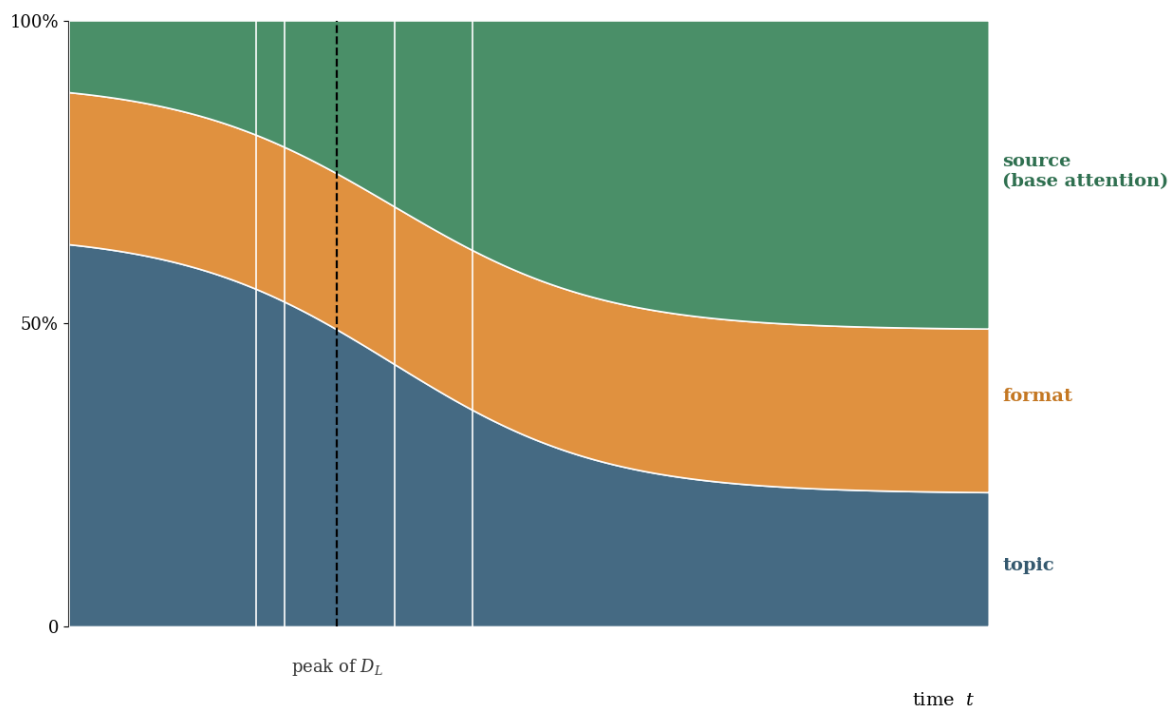
depends on the density of the relationship with the public, the coherence accumulated over time, the legitimacy to address particular topics, and the capacity to activate specific interpretive communities. There are contexts in which the source weighs more than the format; there are others in which the format weighs more than the source; there are also situations in which the topic imposes itself with such force that it demotes, even if provisionally, the relative importance of the other components. The proposed model does not require a fixed and absolute hierarchy among these elements in each and every circumstance. It requires only that their concrete articulation be recognized, along with the fact that, in structural terms, the topic tends to concentrate the greatest variability of demand.

Consequently, the strategic analysis of communication cannot limit itself to the choice of source or format. It is indispensable to understand at what stage of the demand curve for a given topic the content is being positioned, under penalty of investing in strategies that, even if correct from a conceptual standpoint, will be suboptimal from the standpoint of communicational efficiency. An adherent format, a legitimized source, and good technical execution do not suffice when the publication enters a phase of already saturated or declining demand. Likewise, average sources or less prestigious formats can perform above expectations when they position themselves precisely in phases of strong thematic acceleration. The usefulness of the model appears precisely at this point, by shifting the analyst's attention to the concrete articulation among the three components, without losing sight of the relative centrality of the topic.

The interaction among topic, format, and source configures the dynamic triad of demand. These components act interdependently, with weights that vary according to the context and the stage of the thematic curve (Figure 1). Digital performance, therefore, does not derive from the isolated quality of a piece, but from the combination of this triad within an ecology of specific audiences. By integrating these dimensions into a common analytical structure, the model lays the foundations for the introduction of its most critical variable: the movement of the demand curve in time and the application of its derivative to the definition of the optimal timing of publication.

FIGURE 1

Variable weight of the triad (topic, format, source) along the curve



Note. Authors' own elaboration (2026).

## SECTION 4

# The derivative of the demand curve as an indicator of optimal timing

## 4.1 Notation

It is convenient to fix the notation used from here on. Let  $D_L(t)$  be the level of latent demand for a given topic at instant  $t$ , approximated empirically by the proxies discussed in section 2.2.  $D_L'(t)$  is defined as the first derivative of  $D_L(t)$ , that is, the rate of variation of demand in time, and  $D_L''(t)$  as the second derivative, that is, the variation of the growth rate itself. Thus,  $D_L'(t) > 0$  indicates growing demand,  $D_L'(t) < 0$  indicates decline, and the sign of  $D_L''(t)$  informs whether this movement accelerates ( $D_L''(t) > 0$ ) or loses momentum ( $D_L''(t) < 0$ ).

In graphical terms, one can imagine that demand for a given topic over time behaves as a curve that rises, reaches a peak, and eventually enters decline. The first derivative indicates the direction and immediate intensity of this movement; the second derivative informs whether the movement gains strength or loses momentum. The analogy of driving a vehicle helps fix the distinction: speed says how fast one advances; acceleration says whether that pace is increasing or decreasing. The two indicators together describe the movement more precisely than either one alone.

## 4.2 Four phases and five segments of the curve

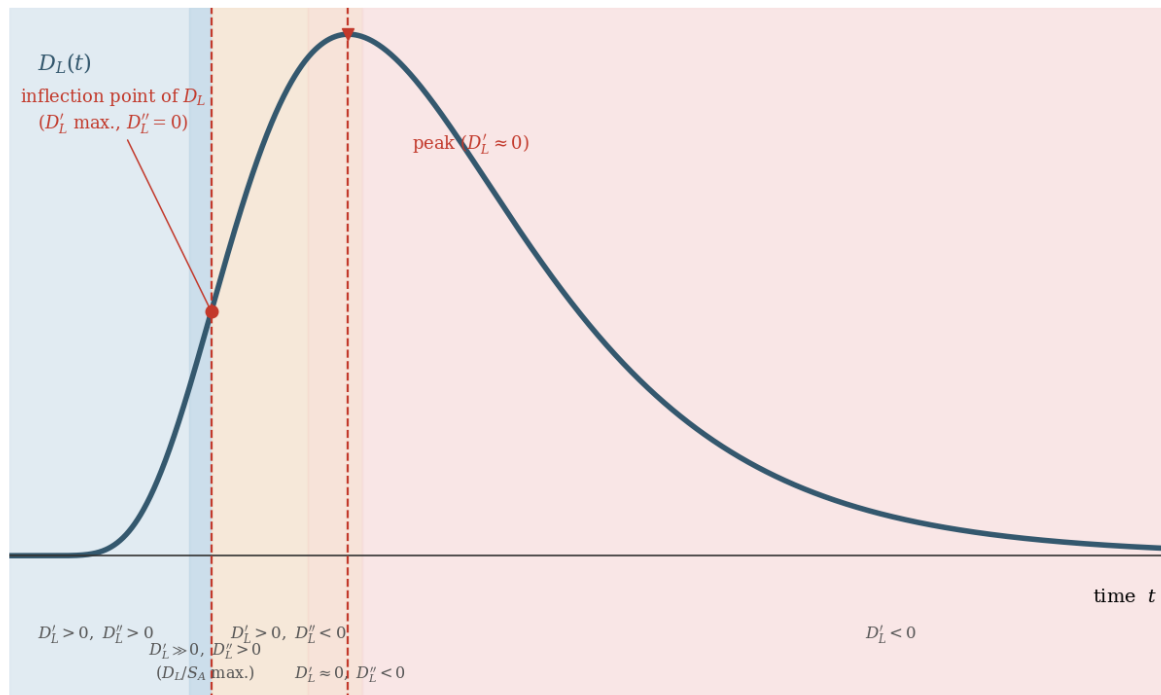
The combination of the signs of the two derivatives distinguishes four distinct strategic situations:

- (i)  $D_L'(t) > 0$  and  $D_L''(t) > 0$ : demand grows and growth accelerates. Open window with strong expansion; the phase of elevated  $D_L''(t)$  signals the approach of the point of maximum communicational efficiency, whose precise definition is given by the ratio  $D_L/S_A$  (sections 2.6 and 4.4).
- (ii)  $D_L'(t) > 0$  and  $D_L''(t) < 0$ : demand still grows, but with loss of momentum. A favorable window, though less promising.
- (iii)  $D_L'(t) \approx 0$  and  $D_L''(t) \leq 0$ : the curve approaches the peak or enters stabilization. Decreasing marginal return; high competition; active supply at an elevated level.
- (iv)  $D_L'(t) < 0$ : demand operates in decline. A progressively less efficient environment for new publications.

For operational purposes, it is useful to refine this taxonomy into five segments, identified by the letters a–e in Figure 2 and in Table 1 (section 7.3). Segment a corresponds to the beginning of phase (i), with growth still incipient and low active supply; segment b, to the mature portion of the same phase, in which  $D_L'$  is high and  $D_L''$  remains positive — the intense acceleration; segment c corresponds to phase (ii); segment d, to phase (iii); and segment e, to phase (iv). The distinction between a and b is one of magnitude, not of sign: both exhibit positive derivatives, but it is typically in b that the ratio  $D_L/S_A$  culminates. The four phases formalize, on the demand side, a pattern described qualitatively by the issue-attention cycle (Downs, 1972), in which topics emerge, capture the center of the debate, and decline without the underlying problem being resolved; the model's contribution lies in coupling to this cycle the cumulative supply side and a decision rule — not merely describing the phases, but indicating what to do in each one.

FIGURE 2

Latent demand curve and the five segments (a–e)



Note. Authors' own elaboration (2026).

### 4.3 Why the peak is not the best moment

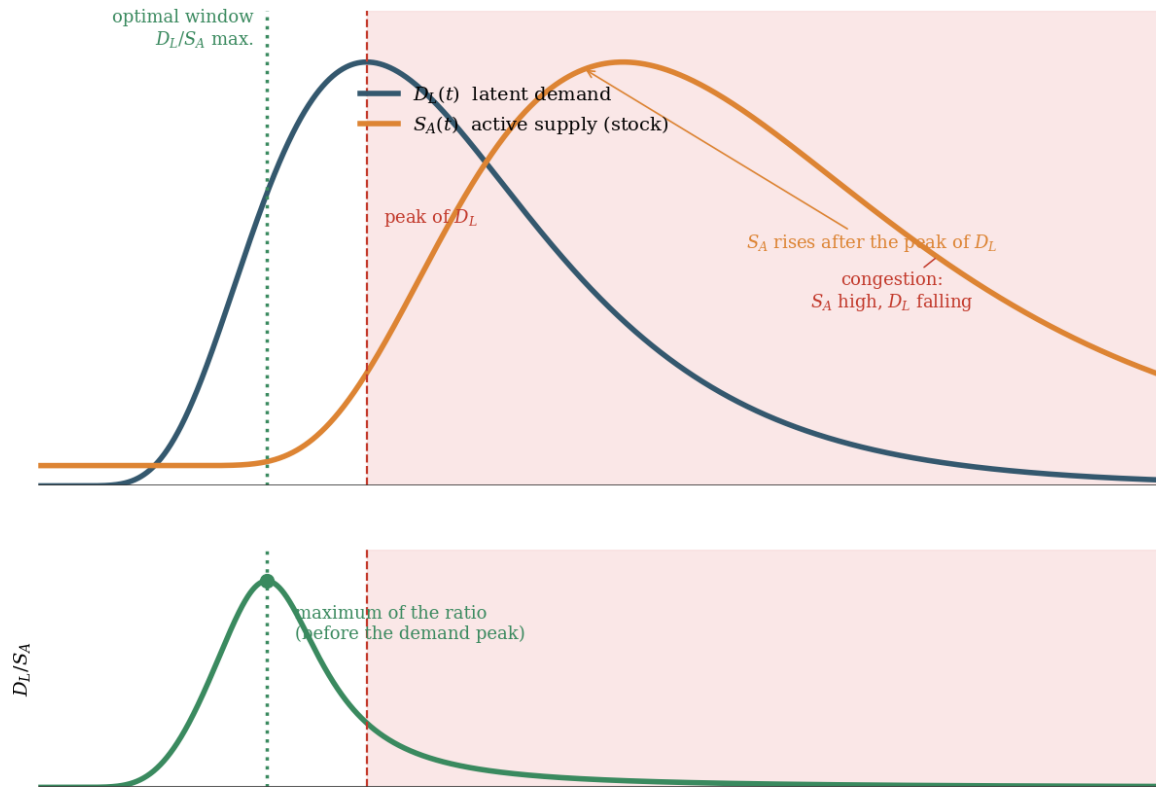
Common intuition suggests publishing when the topic is “at its height”. The model shows that this intuition is imprecise. When the topic reaches the absolute peak of interest, the acceleration that propelled it has already disappeared and, in many cases, active supply has already expanded significantly – not only through the entry of new publications at that instant, but through the accumulation of everything published in the preceding days that remains in algorithmic circulation. The absolute peak of the curve is visible to everyone, which tends to attract more producers, intensify competition, and reduce the marginal yield of each new piece.

The most advantageous window usually lies earlier – in the segment in which demand has already gained enough traction to produce strong thematic valuation, but active supply has not yet kept pace with that growth. In terms of the relational rule of section 2.6, it is the point at which the ratio  $D_L(t) / S_A(t)$  reaches its maximum (Figure 3). To locate it in practice, what matters is to observe jointly the first and second derivatives of the demand curve: the optimal point usually occurs when  $D_L'(t)$  is positive and high,  $D_L''(t)$  is still positive, and active supply has not yet reached the level at which it grows faster than demand. The derivatives indicate where the window is opening; the ratio between demand and supply confirms whether it is still open.

This formulation prevents the analysis of timing from remaining captive to overly intuitive categories such as “trending topic” or “subject of the moment”. These expressions may describe the general state of a topic, but they offer insufficient information about the exact position of the curve or the speed at which its conditions are changing, and they entirely ignore the dynamics of competing active supply.

FIGURE 3

Optimal window:  $D_L / S_A$  maximum before the demand peak



Note. Authors' own elaboration (2026). The figures in this paper are schematic (numerical example) and do not represent empirical data: they illustrate the qualitative relations postulated by the model. In Figures 2 and 3, latent demand is given by an example function — a log-normal curve, which rises with curvature, reaches a rounded maximum, and decays slowly. Active supply  $S_A(t)$  is constructed from published supply  $S_P(t)$  — here taken as lagged demand —, accumulated and weighted by a decay kernel. To draw the figures, a specific form of this kernel was adopted — exponential, with a half-life fixed solely for visualization purposes —, although the model, in the body of the text, does not depend on this choice: all that is required of the kernel is that it equal 1 at publication and decrease with age. The parameters are illustrative, chosen only to make visible the properties the model asserts: the optimal window (maximum of  $D_L/S_A$ ) precedes the demand peak, and active supply, being a cumulative stock, continues to grow even after that peak. Segments a–e follow Table 1. No conclusion of the paper depends on the functional form or the values adopted.

#### 4.4 The window-closure condition

The relational rule permits deriving the optimal instant formally. With the expected reach per piece proportional to the ratio  $R(t) = D_L(t)/S_A(t)$ , with positive and differentiable curves, the first-order condition for an interior maximum of  $R$  is  $R'(t^*) = 0$ , which is equivalent to

$$D_L'(t^*) / D_L(t^*) = S_A'(t^*) / S_A(t^*),$$

where:

$t^*$  — optimal instant of publication: the moment at which the ratio  $R(t)$  reaches its maximum value;

$D_L'(t^*)/D_L(t^*)$  — proportional growth rate of latent demand: how much interest in the topic grows, in percentage terms, per unit of time;

$S_A'(t^*)/S_A(t^*)$  — proportional growth rate of active supply: how much the competing stock grows, in percentage terms, per unit of time;

$R'(t^*) = 0$  — first-order condition: at the highest point of a smooth curve, the derivative is zero — the ratio stops rising before it starts to fall.

In words: the window reaches its optimum — and begins to close — exactly when active supply starts to grow, in percentage terms, as fast as latent demand. Before  $t^*$ , demand grows proportionally faster than the competing stock, and each instant of waiting improves the ratio; afterwards, the relation inverts and each instant of waiting deteriorates it.

The condition makes demonstrable, and not merely plausible, the central thesis of section 4.3. At the demand peak,  $D_L' = 0$ ; if at that instant active supply is still growing ( $S_A' > 0$ ) — which the cumulativeness of the stock makes typical, since the recent flow of publications continues to feed  $S_A$  —, then the ratio is necessarily falling at the peak, and the interior maximum of  $R$ , when the window does open, occurs before it. The peak of public visibility and the point of maximum communicational efficiency do not coincide by construction, whenever the competing stock is still expanding at the moment of greatest interest.

Under the exponential kernel used in the figures,  $w(a) = e^{-\lambda a}$ , the stock dynamics assume a simple form:

$$S_A'(t) = S_P(t) - \lambda S_A(t),$$

where:

$\lambda$  (lambda) — decay rate: the higher it is, the faster content loses circulation; it relates to the half-life through  $\lambda = \ln 2 / t_{1/2}$ ;

$S_P(t)$  — stock inflow: the flow of new publications feeding it at each instant;

$\lambda S_A(t)$  — stock outflow: the share of circulating content that ceases to receive distribution at each instant.

In words: the competing stock grows when more new content enters than exits through decay, and shrinks in the opposite case. Substituted into the closure condition, this dynamic yields  $D_L'/D_L = S_P/S_A - \lambda$ : the window closes when the flow of competing publications, in proportion to the already circulating stock, exceeds the sum of the demand growth rate and the decay rate. In numerical explorations of this configuration, the exact position of  $t^*$  proves sensitive to the response speed of competing supply and to the volume of background supply — slower competition and a smaller background stock push the window farther from the peak —, which indicates that the distance between the optimal window and the peak is itself a strategic variable to be estimated in each topic market, and not a universal constant. The operationalization of these quantities is discussed in Appendix A.

## 4.5 Identifying phases in real time

The application of the preceding conditions presupposes that the agent knows, in real time, in which segment of the curve it finds itself — and this is, in practice, the most difficult step.  $D_L$  is not directly observed: it is es-

timated through partial, noisy, and lagged proxies (section 2.2), and the second derivative of a noisy series is unstable — small measurement oscillations produce spurious sign switches. Four operational precautions reduce the risk of phase error. First, smooth the series before differentiating, with moving windows short enough not to erase the inflection one wants to detect. Second, require convergence of multiple independent proxies — searches, mentions, coverage — before declaring a phase change; isolated signals produce false positives. Third, exploit antecedent calendar signals: a relevant share of the curves is anchored in predictable events — conferences, votes, commemorative dates —, for which the acceleration phase can be anticipated without sophisticated estimation. Fourth, take into account the asymmetry of losses: entering too early costs part of the yield of one piece; missing the window compromises the efficiency of the entire campaign. The asymmetry follows from the stock character of the published piece (section 2.6): published early, it remains active and captures the subsequent acceleration of demand; published late, it does not recover the window already closed. Under uncertainty, the error of anticipation tends to be the less costly one — which reinforces, by another route, the recommendation to shift the decision to before the trigger, developed in sections 5 and 6.

#### 4.6 Case 1: per-piece contracting and the predictable event

The logic developed so far can be illustrated by a hypothetical situation that synthesizes the typical decisions of organizations that contract influencer campaigns around events of public interest. The numbers used are rounded and illustrative, intended to fix the structure of the reasoning, not to represent empirical data.

**Case 1 — Per-piece contracting around a predictable event.** Consider an event of public interest with a known date — an intergovernmental conference, a decisive legislative vote, a session of a multilateral body, or an institutional commemorative date. In the weeks preceding the event, latent demand for the corresponding topic begins to heat up; during the event, it reaches its maximum point; after the event, it declines rapidly. An organization decides to contract a mobilization campaign involving ten influencers, paying R\$ 5,000 per contracted piece, totaling a budget of R\$ 50,000. The typical administrative decision schedules the publication of all pieces during the event, at the moment when the aggregate public visibility of the topic is at its maximum. It so happens that, at that same instant, competing active supply also reaches its maximum: the press covers the event, spontaneous creators publish, experts take positions, institutional profiles speak out, and the cumulative stock of content published in the preceding weeks remains in algorithmic circulation. The aggregate reach of the topic in the ecosystem is high on that day. But the average reach per contracted piece — the variable that effectively matters to the campaign, given per-piece payment — is compressed by the density of the competition. In illustrative values: each piece may capture, on average, on the order of 50,000 views at the peak, for a total campaign reach of close to 500,000. Now consider an alternative scenario in which the same campaign enters circulation a few weeks before the event, in the phase in which  $D_L'(t) > 0$  and  $D_L''(t) > 0$ , when demand is already heating up but competing active supply has not yet kept pace. In illustrative values: each piece may capture, on average, on the order of 120,000 views, for a total campaign reach of close to 1,200,000. Same budget, same influencers, same production quality: the average reach per piece more than doubles in the acceleration phase relative to the peak phase. The difference does not come from the campaign's attributes, but from its position on the joint trajectory of latent demand and active supply.

The case illustrates three points of the model. First, the objective function of a campaign contracted on a per-piece basis (average reach per piece) differs from the function that would measure the health of public debate (aggregate reach of the topic), and this difference changes the calculation of optimal timing. Second, the peak of public visibility and the point of maximum communicational efficiency do not coincide — in general, the second precedes the first. Third, the relevant comparison is not between publishing or not publishing on the topic, but between different moments of publication within the curve, with significantly distinct effects on average yield. To this may be added a reinforcement on the cost side, which the case does not quantify and which remains here as a plausible supposition: influencer fees and contracting costs tend to be procyclical to the curve — hiring influencers when the topic is already the subject of the moment usually costs more. Entry in the acceleration phase thus tends to be doubly advantageous: higher expected reach per piece and lower contracted unit cost.

#### 4.7 Linking timing to production time

The analysis of the derivatives only gains full strategic usefulness when combined with the concrete duration of the production and publication process. Between the decision, the preparation of the briefing, the production of the material, the narrative adjustments, the editing, and the publication, a period elapses that can vary from a few hours to several weeks. If this reaction time is not considered in planning, the agent runs the risk of starting the process when demand is growing, but publishing only when the curve has stabilized or begun to fall, incurring the temporal lag typical of structural-lag models.

Optimal timing is never an abstract instant, independent of the operational conditions of the publishing agent. It always results from the encounter of two temporalities: the external one, of demand, which moves in the social environment, and the internal one, of the production process, which defines the possible speed of response. When these two temporalities fall out of alignment, the publication arrives late even if the thematic reading was correct. Articulated with reaction time, this reading reveals a strong path dependence in the digital ecosystem (Arthur, 1989). Early entries, made while thematic acceleration is still underway, tend to shape the competition and capture the efficiency peak. Late entries — even with similar resources — tend to encounter an environment already transformed and saturated, with eroded marginal yield.

## SECTION 5

# Structural asymmetries and reaction time

## 5.1 Heterogeneity among agents

Different agents face unequal positions in the dispute for attention. Although all act in the same market, they do not operate with the same speed of adjustment, with the same degree of decision-making autonomy, or with the same level of flexibility to convert the perception of opportunity into effective publication. Competition for attention occurs not only among content items, topics, formats, and sources, but also among distinct operational structures, each with its own reaction time.

These temporal differences are articulated with broader inequalities of economic capital, symbolic capital, production infrastructure, public legitimacy, and distribution capacity. Some agents enter the dispute with larger teams, privileged access to data, greater capacity to test formats, and more robust means of sustaining circulation. Others operate with scarce resources, less predictability of reach, and much greater dependence on precise timing calls. The digital content market is, therefore, competitive, but deeply unequal.

## 5.2 Decomposing reaction time

Reaction time, denoted  $\tau$ , is the interval between the instant  $t_0$  at which the agent identifies a window of opportunity and the instant  $t_0 + \tau$  at which the content is published. This interval can be decomposed into two analytically distinct components:

$$\tau = \tau_{decision} + \tau_{production},$$

where:

$\tau$  (tau) — total reaction time: everything that elapses between perceiving the opportunity and publishing;

$\tau_{decision}$  — deliberative component: the time spent defining what to do — allocating budget, choosing sources, aligning core messages, validating internally, contracting suppliers, approving framings;

$\tau_{production}$  — executive component: the time spent making the piece once decided — recording, editing, narrative adjustment, revision, effective publication.

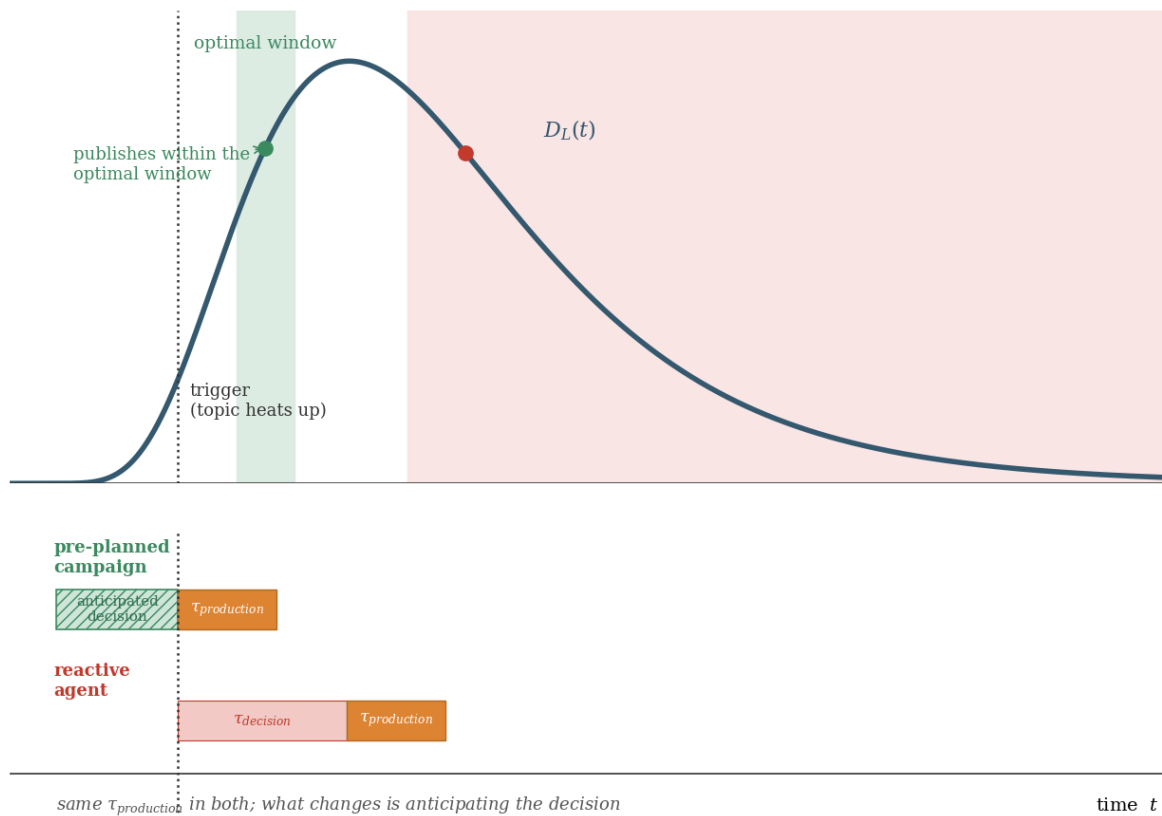
In words: an organization's total delay is the sum of the time it takes to decide and the time it takes to produce — and the two components behave in very different ways. Figure 4 illustrates the decomposition.

The two components are reduced through different mechanisms. Production time has a minimum limit given by the quality the piece requires and by the characteristics of the chosen format — below a certain threshold, the reduction compromises the result. Decision time, by contrast, is structurally compressible and can, to a large extent, be anticipated. This anticipation is the main mechanism through which organizations reduce their total reaction time without compromising the quality of contextualized production.

For a publication to capture the optimal efficiency window, the interval between the perception of the opportunity and the publication must fit within that window. When this condition fails, the publication arrives in a later phase of the curve, with lower marginal return. The reduction of  $\tau_{decision}$ , through prior planning, significantly enlarges the set of windows capturable by the organization — an effect developed in sections 6.5 and 6.6, in the treatment of hybrid curves and campaigns in a state of readiness.

FIGURE 4

Reaction time: decomposition into decision and production



Note. Authors' own elaboration (2026).

### 5.3 The advantage of speed over scale

In environments in which the demand curve for a topic can accelerate and change in hours or a few days, small differences in speed produce very expressive effects on the marginal return of each publication — all the more so as collective attention windows become shorter (Lorenz-Spreen et al., 2019). The agent that reacts early enters the flow when demand is still expanding strongly and when active supply, though already existent, has not yet reached severe levels of saturation. The agent that reacts late enters a market already transformed by the prior circulation of content, at a moment when the relative visibility of each new piece tends to be lower.

As a consequence of this temporal competition, slow agents tend to insert their materials when the ecosystem is already approaching stabilization or decline, while fast sources more frequently capture the window of maximum  $D_L/S_A$  ratio, signaled by the derivatives of the curve (section 4). This dynamic helps explain a frequent paradox of the digital market: simple, spontaneous content often achieves performance superior to that

of complex, highly elaborate pieces — not through intrinsic superiority, but through advantageous temporal location. The pattern is consistent with the experimental evidence discussed in 2.3: if dynamics of social reinforcement amplify small initial differences into highly unequal success trajectories (Salganik et al., 2006), the temporal position of entry — which defines which content items accumulate the first signals of attention — becomes a decisive variable. It should be noted that the temporal extension is an inference of the model, not a result of that experiment. Performance evaluations that dwell only on the creativity or charisma of the source ignore that delay tends to strongly reduce the marginal efficiency of the content: comparing the reach of two publications without considering the position of each one on the demand curve amounts to comparing returns obtained in distinct markets.

The accelerated entry of organic production into the ecosystem generates rapid saturation of active supply, acting as a negative externality for slower actors. The structural penalty imposed on institutional content when it is finally published should not be read, in many cases, as an occasional management failure, but as a reflection of the very organizational architecture of these agents: by choosing to preserve rigor, control, and formal consistency without investing in the reduction of  $\tau_{decision}$ , they tend to sacrifice speed of adjustment and to forgo a central competitive advantage in a market governed by short valuation windows. As discussed in section 4.7, there is in this mechanism a strong path dependence: early entries reconfigure the environment for subsequent entries, so that delay not only reduces the isolated performance of a piece, but displaces the agent into an ecosystem already transformed by competitors' prior action.

The empirical status of the speed premise must, however, be qualified. The claim that smaller, decentralized profiles react faster than institutional agents is a behavioral premise of the model, plausible but not yet demonstrated in this specific arena — and the available external evidence is mixed: in the classic news cycle, professional media were measured leading attention, with individual blogs following at a typical lag of two and a half hours (Leskovec et al., 2009). The direction of the speed asymmetry between types of agent is, therefore, an open empirical question, converted into a hypothesis in section 8 (H2).

The strategic implication is broad. Agents that operate more slowly need to recognize that their main disadvantage lies not only in budget limitations or in the smaller reach of the source used, but in the lag between the movement of demand and their own response cycle. In many cases, the central issue is not to produce more, nor to produce better in an abstract sense, but to reorganize internal processes so as to reduce  $\tau_{decision}$ . The strategic gain may lie less in increasing resources and more in shortening lags through prior planning.

This reading also helps explain why the digital content market so frequently favors light, adaptive, decentralized agents. In environments of high thematic volatility, competitive advantage tends to shift toward those able to operate close to the acceleration of demand. This does not eliminate the power of large structures, nor does it render irrelevant the strength of consolidated sources. It merely indicates that, in the initial phases of a topic's valuation, speed weighs disproportionately. In later phases, when demand stabilizes or declines, other assets gain relative importance — especially the source's base attention and the capacity to access niches not yet exhausted. It is precisely this regime change that opens the way for the following discussion of topic saturation, the source's base attention, and hybrid curves.

## SECTION 6

# Topic saturation, base attention, and hybrid curves

## 6.1 When the topic saturates

When a topic has already been widely explored by multiple actors and demand is stabilizing or declining, the potential for new content to generate high reach comes to depend less on interest in the topic itself and more on the source's capacity to attract attention regardless of the subject addressed.

Saturation does not stem only from the quantitative increase in active supply. It also results from the repetition of framings, the loss of narrative novelty, informational redundancy, and the fatigue produced by repeated exposure to the same type of treatment of the topic. A topic can remain strong in aggregate terms and still begin to yield less because its framings have become predictable. In certain cases, saturation also takes the form of affective fatigue, when repeated exposure to content of high emotional charge reduces the public's disposition to keep reacting with the same intensity. This phenomenon converses with the perspective of an ecology of attention, in which the public is not a homogeneous quantity to be maximized, but a finite cognitive environment in which different forms of exposure produce distinct regimes of receptivity, repetition, and wear (Citton, 2017).

## 6.2 The source's base attention

In saturated environments, sources with a high degree of public recognition, strong adherence to a specific community, or a relatively loyal base tend to keep obtaining above-average results, even when the aggregate performance of content on that topic is falling. In these cases, what sustains performance is no longer the expansive force of demand for the topic, but the existence of a continuous demand for the source itself — an audience interested in following it even when the topic's temperature is no longer the same.

The source's base attention is neither a fixed nor a homogeneous attribute. It varies with reputation, recurrence, communicative style, the density of the bond with the audience, the legitimacy to address certain topics, and the degree of cohesion of the niche around that profile or institution. In some cases, it is sufficient to sustain high results even when the topic is in clear saturation; in others, it functions only as a mechanism of partial cushioning of the fall. As the thematic curve matures, the composition of performance changes: the relative weight of the topic tends to decrease and the relative weight of the source tends to grow.

## 6.3 The internal exhaustion curve

Each source has an internal saturation curve with respect to the same topic, denoted  $E_i(t)$ . After a certain number of posts on the same story, the audience shows signs of tiredness, reflected in drops in views, likes, shares, comments, retention, and other forms of response. The effective performance of a piece therefore depends on an articulation among the general demand curve for the topic, competing active supply, the internal exhaustion curve of the publishing source, the niche that source addresses, and the format adopted.

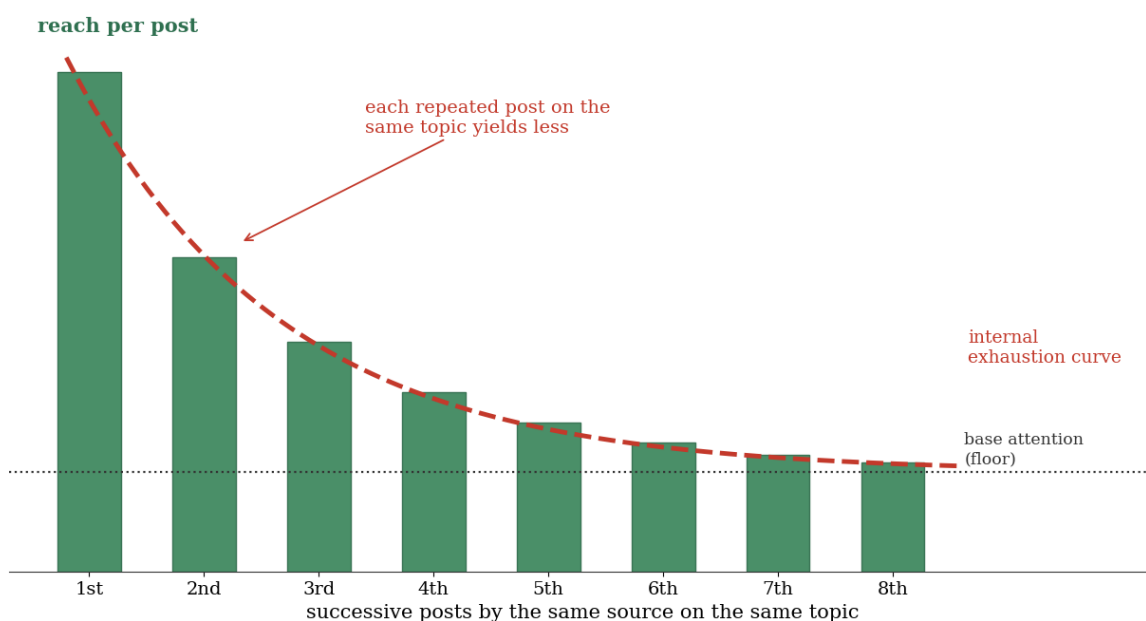
In the initial and intermediate phases of the macro curve, latent demand for the topic tends to dominate — the strength of the topic explains most of the variation in performance. In late phases, the internal exhaus-

tion curve and the source's niche come to dominate — the prior relationship between source and audience sustains results that the topic, by itself, would no longer sustain. This alternation of dominance is formalized in section 7.2.

The internal exhaustion curve follows, in many cases, a pattern similar to that of diminishing returns. The first pieces published on a topic by a given source encounter strong receptivity, especially when there is adherence among topic, niche, and that profile's position. Subsequent publications continue to perform well, but at a gradually lower rate. After a certain point, repetition reduces perceived novelty, compresses the audience's disposition to keep consuming the same framing, and makes the marginal return of new pieces decline rapidly (Figure 5). The pattern is consistent with the advertising-repetition literature, which documents an initially increasing and then declining audience response to repeated exposure — wearout (Schmidt & Eisend, 2015). The source's strength encounters limits of its own: it can cushion the effects of macro-thematic saturation, but does not eliminate them.

FIGURE 5

Source exhaustion: diminishing returns per publication



Note. Authors' own elaboration (2026).

## 6.4 Macro and micro: two articulated levels

The analysis must articulate two distinct but inseparable levels. There is a macro level, relative to the demand curve for the topic in the broader ecosystem, and a micro level, relative to the disposition of each source's audience to keep consuming content on the same theme. Strategies that ignore the micro level overload particular profiles with repetitive content, weakening their capacity to convert attention even when there is still traction at the general level. Strategies that ignore the macro level attribute excessive weight to the source's strength and neglect the fact that the thematic curve may be operating under much less favorable conditions. The correct reading requires crossing the two scales.

Each profile, institution, outlet, or creator organizes around itself a micro-structure of demand — an interpretive community with its own repertoires, specific tolerances of repetition, and particular rhythms of thematic exhaustion. Saturating a source frequently means saturating part of the niche it addresses, since that audience tends to be repeatedly exposed to the same topic by the same voice, the same framing, and similar narrative forms. Micro-thematic saturation thus occurs analogously to the macro kind, but on a reduced scale and with highly relevant effects on the individual performance of the content. The niche must be treated as an inseparable component of the source, because it is there that the internal curves of interest, fatigue, and response take shape.

## 6.5 Hybrid curves and campaigns in a state of readiness

There is an advanced way of operating in environments of saturation or decline: combining the focal topic with a secondary topic that functions as a hook. This second topic must have its own demand curve, ideally on the rise. The technique consists in crossing a topic whose curve has already lost strength with another whose trajectory is accelerating, producing a composition that temporarily reopens more favorable circulation conditions. Instead of insisting on the isolated trajectory of a saturated topic, the source reinserts it into a new conjuncture of interest through articulation with another, hotter topic.

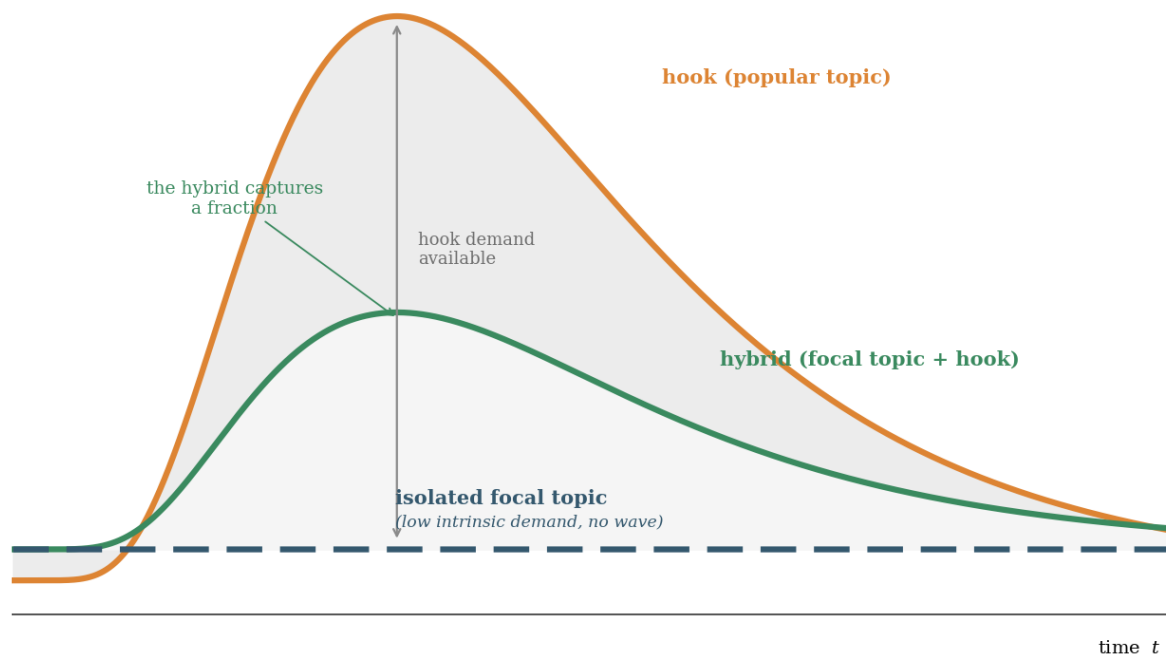
This operation configures a hybrid demand curve: by narratively coupling the saturated topic to a rising topic, the content comes to access part of the second topic's latent demand not yet captured by active supply (Figure 6). The strength of the hook lies in transferring part of the energy of the ascending curve to the weakened topic, renewing its power to capture attention through a new key of entry. The effective gain depends on two factors: the intensity of the trigger topic's ascending curve and the substantive adherence of the articulation between the two topics. When this articulation is artificial or excessively opportunistic, the reach gain tends to be small; when it is well constructed, the hybrid curve can reopen a significant valuation window.

Hybrid curves show that saturation does not imply the absolute exhaustion of circulation capacity. What is exhausted is a particular autonomous trajectory of thematic valuation. There may still exist forms of repositioning, re-presentation, or reframing that set the topic in motion again by connecting it to another curve of interest. This type of operation requires a fine reading of the environment, sensitivity to identify consistent hooks, and the capacity to act before the new topic also enters its own saturation phase. Reaction time reappears, therefore, as a decisive variable: a thematic connection made early can reopen a content item's valuation; the same connection, made late, can sound redundant and lose a large part of its potential.

The operationalization of hybrid curves is particularly demanding for topics structurally weak in autonomous volatility — topics relevant in a broad sense, but which rarely trend on their own. For these cases, the efficient strategy requires a specific reorganization of reaction time, described below.

FIGURE 6

Hybrid curve: the focal topic accesses part of the hook's demand



Note. Authors' own elaboration (2026).

## 6.6 Case 2: campaigns in a state of readiness

The combination of hybrid curves, the decomposition of reaction time, and unpredictable triggers can be illustrated by a second hypothetical situation, complementary to the previous one.

**Case 2 — Topic of low autonomous volatility and reactive trigger.** Consider an organization that intends to communicate on a structurally relevant topic of low autonomous volatility — a topic that rarely trends on its own, but that can be coupled to the ascending curves of other topics. External events potentially capable of functioning as triggers — a localized environmental disaster, an unexpected political crisis, an abrupt regulatory change, a judicial decision of broad scope — occur at unpredictable moments. When the trigger appears, the favorable hybridization window is short: the first and second derivatives of the trigger topic's curve are positive and high, but competing active supply grows rapidly. Capturing this window requires that the total time between identification of the trigger and effective publication fit within a narrow interval. An organization that begins the entire process only after the trigger — defining the budget, choosing influencers, contracting, aligning the framing, producing, publishing — incurs a high  $\tau_{decision}$ . By the time the piece comes out, the window has already closed: competing active supply has saturated the environment. The efficient strategy shifts  $\tau_{decision}$  to before the trigger. The organization plans the campaign in advance: it defines the focal topic, contracts the influencers on standby, establishes criteria for activation, maps possible framings and core messages. The piece itself is not produced in advance, because producing before the concrete trigger would compromise adherence to the specific case that will activate it. What is ready beforehand is the architecture of the campaign, not the finished content. When the trigger appears,  $\tau_{decision}$  is practically zero. What remains is only  $\tau_{production}$  — the contextualized production of the piece that speaks to the concrete case that opened the window. Effective reaction time falls significantly, and the publication has a real chance

of occupying the phase of maximum efficiency of the hybrid curve. The result is that the efficient strategy for topics of low autonomous volatility requires less investment in fast production and more investment in prior planning. The campaign is not designed to be published on a set date — it is designed to be published when the right curve appears. In economic terms, it is an architecture for the rapid exercise of a decision that could, in other circumstances, take weeks.

The case illustrates three points of the model. First, the reduction of reaction time is primarily a reduction of  $\tau_{decision}$ , not of  $\tau_{production}$ : producing before the trigger compromises adherence; planning before the trigger does not. Second, topics structurally weak in autonomous volatility do not depend on a publication calendar, but on an architecture of readiness that permits contingent activation. Third, in configurations of this type, a larger budget and more elaborate production still matter, but their return is strongly dependent on anticipated planning: campaigns with more resources but no rapid-response capacity tend to yield less than more modest campaigns with  $\tau_{decision}$  at zero at the critical moment.

## SECTION 7

# Strategic implications and intertemporal allocation

The formulations developed so far permit extracting more general strategic implications for resource allocation in the digital content market. If performance depends on the relative position the content occupies on the demand curve, on the behavior of its derivatives, on the simultaneous expansion of active supply, on the agent's reaction time, and on the source's base attention in contexts of saturation, then the decision about where to invest can no longer be guided solely by abstract preferences of language, generic intuitions about engagement, or static criteria of reputation. Efficient allocation requires reading the stage of the curve and matching the instruments of communication to the specific conditions of thematic valuation present at each moment.

## 7.1 Guidance by stage of the curve

In the acceleration phases — segments a and b of the curve, in which  $D_L''(t)$  remains positive and the ratio  $D_L/S_A$  is high or culminating —, the main engine of performance lies in the topic, and not in the intrinsic strength of consolidated sources. As developed in section 4.3, it is the phase of highest marginal return per piece; in allocative terms, it becomes rational to concentrate resources on the production of content able to enter the ascending flow of demand rapidly, prioritizing speed, thematic adherence, and the capacity to multiply presence in the ecosystem. Efficiency, at this stage, does not depend so much on resorting to the most expensive or most prestigious sources: it depends on inserting relevant content into a still favorable conjuncture, before the dispute for attention thickens — which includes investing in production capable of reacting with precision and publishing within the window.

When growth loses momentum (segment c, with  $D_L''(t) < 0$ ) and, later, when the curve enters a plateau (segment d, with  $D_L'(t) \approx 0$ ), the environment changes in nature. Active supply tends to expand faster than interest, competition thickens, and mere presence in the flow ceases to guarantee expressive returns. This moment calls for a more selective combination of thematic choice, format, and source: there is still room for good results, but the cost of late entry grows, and the dispute comes to demand stronger differentiation — by framing, by niche, or by the source's strength.

In the decline phase (segment e, with  $D_L'(t) < 0$ ), maintaining high publication volumes tends to produce an inefficient use of resources: additional content is launched into an environment of low relative demand and high accumulated saturation, in a logic typical of diminishing returns. It is at this point that the main engine of performance tends to shift from the topic to the source. In contexts of saturation or decline, base attention gains increasing weight, because the general valuation of the topic no longer sustains, by itself, the reach of new pieces. It becomes more rational to concentrate any remaining resources on sources capable of mobilizing their own attention — especially those that have not yet exhausted the topic on their profiles and retain access to relatively unsaturated niches — and to reduce more firmly the volume of publications strictly anchored in the declining topic.

## 7.2 The central allocation rule

This regime change helps synthesize one of the model's central propositions. When demand for attention grows faster than the active supply of competing content on the topic, the most efficient strategy tends to concentrate primarily on content. In contexts of thematic saturation, the most rational strategy shifts the weight of investment to the source, provided it still has an unexhausted niche and sufficient base attention to sustain circulation above the topic's average. It does not propose a rigid opposition between content and source, nor does it suggest that either ceases to matter at any stage of the curve. What it does is indicate which variable tends to concentrate the greatest leverage in each phase of this dispute for attention.

The rule can be condensed into an expression. The expected reach of a piece by source  $i$  at instant  $t$  is written

$$\text{expected reach of the piece by } i \propto \theta(t) \cdot D_L(t)/S_A(t) + (1 - \theta(t)) \cdot B_i \cdot e_i(t),$$

where:

$\theta(t)$  (theta) — conjunctural weight of the topic at instant  $t$ , with values between 0 and 1: high in the acceleration phases of the thematic curve and declining as it matures;

$D_L(t)/S_A(t)$  — ratio of latent demand to active supply: the attention available per unit of competition (section 2.6);

$B_i$  — base attention of source  $i$ : its stable capacity to mobilize its own audience, regardless of the topic (section 6.2);

$e_i(t)$  — internal exhaustion factor:  $E_i(t)$  normalized to the interval  $[0, 1]$ ; it falls as the accumulated number of the source's pieces on the same topic grows (section 6.3);

$(1 - \theta(t))$  — complementary weight: the share of performance that the topic's conjuncture ceases to explain is explained by the source's own strength.

In words: a piece's reach combines two engines — the topic's conjuncture (first term) and the source's own strength (second term) —, whose weights alternate along the curve. The expression does not purport to be estimable term by term; its role is to render unequivocal the central claim of this section: in the initial phases, the first term dominates and the lever is content positioned early; in the late phases, the second term dominates, and the lever becomes the source with its own base and an unexhausted niche.

This rule must, evidently, be read in articulation with format and niche. In initial or intermediate phases of acceleration, content poorly adherent to the platform's dominant language can waste an important part of the available thematic valuation. In late phases, a strong source already internally saturated with respect to the same topic can yield below expectations even while retaining a large audience. The efficient decision therefore depends on crossing the stage of the curve, the sensitivity of performance to format, the source's base attention, and the degree of exhaustion of the niche it addresses. The general rule remains useful because it organizes the variable hierarchy of these factors over time: the relative weight of the variables is not fixed, but shifts as the trajectory of thematic demand matures, decelerates, and loses strength.

### 7.3 Summary table

Table 1 consolidates the allocative guidance by segment of the curve, bringing together the four phases of section 4.2 in the five operational segments defined there and the implications discussed in 7.1 and 7.2.

TABLE 1

Allocative guidance by segment of the demand curve

| SEGMENT | $D_L'(\tau)$ | $D_L''(\tau)$ | PREDOMINANT DYNAMICS                       | ALLOCATIVE GUIDANCE                                               |
|---------|--------------|---------------|--------------------------------------------|-------------------------------------------------------------------|
| a       | $> 0$        | $> 0$         | Initial growth; low active supply          | Early entry; format testing; traction gains                       |
| b       | $> 0$ high   | $> 0$         | Intense acceleration; $D_L/S_A$ maximum    | Concentration of resources on content and speed                   |
| c       | $> 0$        | $< 0$         | Growth with loss of momentum; supply rises | Greater selectivity; careful combination of topic, format, source |
| d       | $\approx 0$  | $\leq 0$      | Stabilization; high active supply          | Shift toward strong sources, niche, differentiation               |
| e       | $< 0$        | $\pm$         | Decline of thematic demand                 | Reduced exposure; reframing; hooks and hybrid curves              |

Note. Authors' own elaboration (2026).

### 7.4 Reducing reaction time as a strategic priority

As established in section 5.2 and illustrated by Case 2, reducing reaction time — primarily  $\tau_{decision}$  — is decisive whenever the strategy intends to capture acceleration phases. Optimal allocation requires deciding not only how much to invest, but in which structures to invest in order to shorten lags: anticipating the decision through prior planning, shortening approvals, creating rapid-response repertoires, operational arrangements that permit publishing while windows are still favorable. The model thus guides not only the content of the decision; it also guides the temporal architecture of the very capacity to decide.

Another important implication appears in the use of hybrid curves in contexts of saturation. When demand for a main topic has entered deceleration or decline, an advanced way of recovering efficiency consists in articulating it with a secondary topic whose demand curve is rising. In allocative terms, linear insistence on an exhausted topic is not always the most rational option. There are situations in which the best decision is to reframe the topic, reposition it, or connect it to a new ascending curve, provided this articulation preserves intelligibility and public adherence.

In sum, resource allocation in digital environments of scarce attention is a problem of temporal coordination: recognizing the stage of the curve, the behavior of its derivatives, the degree of expansion of active supply, the agent's reaction time, the strength of the source's base attention, and the signs of niche exhaustion — and deciding, dynamically, when to enter, with what type of content, through which source, and when to reduce, redirect, or hybridize the thematic investment.

Finally, a natural extension of the framework should be registered. The entire analysis of this section is intra-topic: it allocates resources along the curve of a single topic. The real allocator frequently operates a portfolio of topics, with lagged and partially correlated curves, in which the opportunity cost of entering one topic is the window missed in another. The multi-topic portfolio problem — simultaneous allocation across competing curves — remains open and constitutes a direct extension of this model.

## SECTION 8

# Limitations of the model and research agenda

Six main limitations deserve attention.

The first concerns the very nature of data on demand. The proxies available for  $D_L(t)$  — searches, mentions, hashtags, coverage intensity — are partial indicators, with structural biases. They underestimate attention mobilized exclusively on closed platforms, overestimate topics that have achieved broad indexing, and are sensitive to the very algorithmic language of the measurement tools. The proxies available for  $D_R(t)$  — reach, engagement, retention — are contaminated by the variables whose influence one wants to understand. Good practice is to triangulate sources, evaluate convergence, and critically discuss the gaps between observed demand and real demand, without expecting a complete resolution of the circularity. To this fragility is added the competitive weighting of  $S_A$ : without an ex ante rule for the weights  $\kappa$  (section 2.4), the ratio  $D_L/S_A$  could accommodate ex post any performance anomaly, which would compromise its falsifiability. Defining and fixing the weighting scheme before any test is, for this reason, a first-order methodological requirement.

The second limitation concerns the algorithmic opacity of the platforms. The model presupposes recommendation systems as active mediators, but has no full access to their distribution criteria. Part of the effects attributed to the relation between supply and demand may derive from specific algorithmic adjustments whose intensity is not publicly measurable. As Nieborg and Poell (2018) argue, the contingent character of content under platformized governance means that its circulation depends on rules that change unilaterally and without prior notice, which structurally limits any predictive pretension of the model. Comparability across platforms (X, Instagram, TikTok, YouTube) is also limited, given that each presents its own distribution logics — including distinct half-lives  $t_{1/2}$ , whose estimation depends on circulation data the platforms do not publish.

The third limitation is the absence of an explicit monetization dimension. The model deals with the economics of attention — the allocation of a scarce cognitive resource — but does not directly model the financial, advertising, or political return of circulation. The articulation between reach obtained and economic value converted (into revenue, sales, votes, sustained mobilization) constitutes an analytically and empirically distinct layer that remains a natural extension of the model.

The fourth limitation is the absence, in this text, of systematic empirical tests. The paper works at a theoretical-analytical level and with preliminary formulations. A future research agenda can be organized around testable hypotheses: (H1) content published in segment b of Table 1 will exhibit higher average reach than that of segments d and e, controlling for source and format; (H2) the difference in reaction time between institutional agents and decentralized agents will be positive and all the more determinant of performance the shorter the acceleration interval — a premise whose direction, as seen in 5.3, the external evidence still disputes; (H3) in saturation phases, performance will correlate more strongly with the source's base attention than with the aggregate volume of searches for the topic; (H4) the introduction of a rising thematic hook will generate, in content on declining topics, a statistically significant reach gain relative to equivalent publications without a hook; (H5) campaigns with anticipated  $\tau_{decision}$  will capture windows in a significantly higher proportion than campaigns with  $\tau_{decision}$  subsequent to the trigger, controlling for budget and production quality; (H6) among the three dimensions of the triad, time series of interest by topic will exhibit variance and oscillation amplitude higher than those of comparable series of format adoption and source attention, in comparable

windows — converting the postulate of section 3.4 into a test; and (H7) in more professionalized topic markets, the average lead time of publications relative to the demand peak will increase over time, as the anticipation rule diffuses among the agents able to exercise it (section 2.7). For H1, H2, and H5, predictable and recurrent events — climate conferences, fixed-calendar votes — offer natural observation panels; for H4, event-study designs around triggers; for H6, public series of searches and of format adoption; for H7, longitudinal series of the interval between publication date and the peak of interest in the topic.

The fifth limitation is the abstraction of a single objective function. The model, by making explicit that campaigns contracted on a per-piece basis maximize average reach per piece, simplifies a reality in which organizations frequently operate with multiple and partially conflicting objectives — reach efficiency, institutional presence, political legitimacy, accountability, sustained mobilization. Extensions of the model can incorporate composite objective functions, in which the relative weight of the components varies according to the type of organization and the context of the campaign.

The sixth limitation is the partial-equilibrium character of the model, made explicit in section 2.7: the recommendations take the behavior of the other agents as given. The generalization of the anticipation rule would displace the windows themselves, and the persistence of the advantage depends on the heterogeneity of reaction times. Game-theoretic extensions — games of timing and preemption — are the natural path for endogenizing this dynamic.

There is, moreover, a normative dimension that the model does not resolve and that its use cannot ignore. The logic of hybrid curves recommends, in the limit, coupling agendas to rising triggers — including crises and tragedies. The boundary between substantive articulation and predatory opportunism is not definable in terms of efficiency: it is an editorial and ethical judgment that the model informs, but does not replace. The same holds for the risk of thematic herd behavior and for the manipulation of the demand proxies themselves as rules of this kind diffuse. Models of attention allocation describe how to capture it efficiently; they do not say what deserves to capture it.

Finally, it is worth observing that the model is especially useful for agents whose social function involves intervening in public disputes for attention. Extensions can explore adaptations to specific contexts — electoral campaigns with fixed windows, crisis communication in exogenous events, science and public health communication, communication in highly polarized environments —, in which the same analytical framework may need to incorporate additional variables.

## SECTION 9

## Concluding remarks

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The model consolidates the circulation of digital content as a problem of intertemporal allocation under scarcity, as an alternative to analyses limited to individual creativity or to algorithmic adjustments. Demand is structured on the conjunctural articulation among topic, format, and source; being the most volatile variable, the topic dictates the abrupt oscillations of performance, while the format governs replicability and the source acts as an anchor of attention in specific niches, with the dominance of each pillar alternating as the thematic curve matures.

The paper's specific contributions are articulated on three planes. On the conceptual plane, the distinction between latent demand and realized demand confronts the methodological circularity of measuring interest in algorithmically mediated systems; the distinction between published supply and active supply recognizes the cumulative character of saturation; and making average reach per contracted piece explicit as the objective of campaigns paid per unit reorganizes the calculation of optimal timing. On the analytical plane, the first and second derivatives of the curve refine the notion of timing; the relational rule synthesizes the optimal position as the point of maximum ratio between latent demand and active supply; the window-closure condition — equality between the proportional growth rates of latent demand and active supply — makes the window's precedence over the peak demonstrable; the decomposition of reaction time into decision and production gives precision to the discussion of lags; and the hybrid curve formalizes the concept of thematic recombination. On the operational plane, the two cases presented — per-piece contracting around a predictable event and a campaign in a state of readiness for a reactive trigger — show how the abstract formulations translate into concrete organizational decisions, with measurable consequences for performance.

The practical usefulness of the model appears on three fronts. For communication managers, it offers a more rigorous language for diagnosing the competitive environment and organizing allocative decisions. For funders and foundations, it offers more consistent criteria for impact evaluation and resource allocation. For researchers, it offers testable hypotheses and an analytical structure for comparative studies.

The model is, however, a point of departure: its consolidation depends on future empirical work, on dialogue with other traditions in the political economy of communication, and on adaptations to specific contexts of application — a task this paper seeks to render more precise and productive.

## APPENDIX A

# Consolidated notation and operationalization

## A.1 Main variables

$D_L(t)$  — latent demand: potential social interest in the topic at instant  $t$ .

$D_R(t)$  — realized demand: the share of latent demand that converts into observable behavior within the ecosystem. By construction,  $D_R(t) \leq D_L(t)$ .

$S_p(t)$  — published supply: flow of new content on the topic entering the system at  $t$ , counted in units of competitive weight  $\kappa$  (section 2.4).

$S_A(t)$  — active supply: stock of content still receiving algorithmic distribution at  $t$ , given by the integral  $S_A(t) = \int_{-\infty}^t S_p(u) w(t-u) du$ , where  $w(a)$  is the decay kernel ( $w(0) = 1$ , decreasing) and the half-life  $t_{1/2}$  satisfies  $w(t_{1/2}) = 1/2$ .

$E_i(t)$  — internal exhaustion curve of source  $i$ 's audience with respect to the topic; it enters the allocation rule (A.5) through the factor  $e_i(t)$ , with values between 0 and 1.

$B_i$  — base attention of source  $i$  (section 6.2).

$\theta(t)$  — conjunctural weight of the topic in the allocation rule (A.5), with values between 0 and 1: high in acceleration, declining in the maturity of the curve.

$\tau$  — reaction time: interval between identification of the window and effective publication, with  $\tau = \tau_{decision} + \tau_{production}$ .

## A.2 Derivatives

$D_L'(t) = dD_L/dt$  — speed of demand.

$D_L''(t) = d^2D_L/dt^2$  — acceleration of demand.

## A.3 Reaction time

$\tau_{decision}$  is the deliberative component of reaction time — allocating budget, choosing sources, aligning core messages, validating internally, contracting suppliers, approving framings. It is structurally compressible through prior planning.  $\tau_{production}$  is the executive component — recording, editing, narrative adjustment, revision, effective publication. It has a minimum limit given by the quality the piece requires and by the characteristics of the format. The strategic reduction of reaction time comes primarily from the anticipation of  $\tau_{decision}$ , without compromising the contextualization of the produced piece.

## A.4 Relational rule and closure condition

$$\text{expected reach per piece} \propto D_L(t) / S_A(t)$$

Reading: the ratio divides the attention potentially available for the topic ( $D_L$ , latent demand) by the effective competition disputing that attention ( $S_A$ , active supply weighted by competitive weight). The higher the ratio at the instant of publication, the higher the piece's expected reach. The optimal instant of publication is the  $t^*$  at which this ratio reaches its maximum, characterized by the first-order condition

$$D_L'(t^*) / D_L(t^*) = S_A'(t^*) / S_A(t^*).$$

Reading: each side is a proportional growth rate (percentage growth per unit of time) — latent demand on the left, active supply on the right. The window closes when active supply starts to grow, in percentage terms, as fast as demand. Whenever  $S_A' > 0$  at the demand peak — the typical case, given the cumulativeness of the stock —,  $t^*$  precedes the peak. The derivatives  $D_L'(t)$  and  $D_L''(t)$  function as anticipatory signals of the opening and closing of this window. Under an exponential kernel,  $w(a) = e^{-\lambda a}$  with  $\lambda = \ln 2 / t_{1/2}$ , it holds that  $S_A'(t) = S_p(t) - \lambda S_A(t)$  — the stock grows when the inflow of new publications exceeds the outflow through decay — and the closure condition becomes  $D_L' / D_L = S_p / S_A - \lambda$ . The generalization  $D_L^\alpha / S_A^\beta$ , with  $\alpha, \beta > 0$  (section 2.6), preserves the existence and the precedence of the window, altering only its exact position.

## A.5 Allocation rule by phase

expected reach of the piece by source  $i \propto \theta(t) \cdot D_L(t) / S_A(t) + (1 - \theta(t)) \cdot B_i \cdot e_i(t)$

Reading: the first term is the conjunctural engine — the ratio of latent demand to active supply, weighted by the topic's weight  $\theta(t)$ ; the second is the source's own engine — base attention  $B_i$  discounted by internal exhaustion  $e_i(t)$ , weighted by the complement  $(1 - \theta(t))$ . In the initial phases of the curve,  $\theta(t)$  is high and the first term dominates: the lever is content positioned early. In the late phases,  $\theta(t)$  falls and the second dominates: the lever is the source with its own base and an unexhausted niche (section 7.2).

## A.6 Objective function

For campaigns that pay a unit price per contracted piece, with the number of pieces and the total budget fixed, the relevant objective function is the maximization of the average reach of the contracted pieces, and not of the aggregate reach of all circulation of the topic in the ecosystem. The decision variables are the timing, the source, the format, and the framing of each piece.

## A.7 Hybrid curves

A hybrid curve results from the substantive articulation between a focal topic (in saturation or decline) and a secondary topic (a hook, on the rise). The potential reach gain depends on the intensity of the hook's ascending curve and on the adherence between the two topics. Artificial or opportunistic articulations produce small gains; intelligible and pertinent articulations can reopen valuation windows for topics structurally weak in autonomous volatility.

## A.8 Operationalization

The decay kernel  $w(a)$  and the half-life  $t_{1/2}$  can be estimated from the decay of daily impressions per piece, in panels of publications tracked from their posting date — a procedure analogous to those used in the literature on the dynamics of collective attention (Candia et al., 2019; Wu & Huberman, 2007). The proxies for  $D_L$  follow the triangulation of section 2.2 (searches, mentions, coverage). The competitive weights  $\kappa$  must be fixed ex ante, by source and format, as section 8 requires. The parameters of the allocation rule of section 7.2 —  $\theta(t)$ ,  $B_i$ , and  $e_i(t)$  — are calibrable through performance regressions on panels of publications with source fixed effects.

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1. The canonical Walrasian model describes markets through the crossing of supply and demand curves at a stable equilibrium point, with price and quantity determined simultaneously. Here, this reference serves only as a counterpoint: the attention market does not operate through static equilibria, but through temporal trajectories in which supply and demand move at distinct rhythms and under nonlinear algorithmic mediation. The most useful tools for describing it come from intertemporal dynamic analysis, from the economics of information (Shapiro & Varian, 1999), from the theory of attention markets (Anderson & de Palma, 2012), and from the literature on path dependence and increasing returns (Arthur, 1989).



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